

End-to-End Energy-Efficiency and Reliability of UAV-Assisted Wireless Data Ferrying

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Abstract—We analyze the end-to-end performance of an unmanned-aerial-vehicle (UAV)-assisted data ferrying network where the UAV serves as a data ferry between the source base station (BS) and multiple destination receivers. We evaluate the end-to-end reliability both with and without packet retransmission technique called automatic repeat request (ARQ), bit error probability (BEP), energy-efficiency, and transmission outage probability. We consider line-of-sight (LoS) and non-line-of-sight (NLoS) transmissions in both the data loading and delivering links and model them with the Rician and Rayleigh fading channels, respectively. We derive tractable approximations for the derived SNR outage results and demonstrate their application in optimizing the distance that a UAV should travel in order to balance the energy-coverage trade-offs. We formulate two different optimization problems and convexify them to solve for the optimal ferrying distance, i.e. (i) outage-constrained energy minimization and (ii) energy-constrained SNR outage minimization. Closed-form optimal solutions are obtained for the second problem. In addition, we formulate a bi-objective optimization problem in order to minimize SNR outage and energy consumption with desired SNR outage probability constraints. The objective function is then reformulated using difference of convex functions (DC) and solved using a DC algorithm. Numerical results validate the derived expressions and show a comparison of the obtained solutions with the solutions obtained from exhaustive search. Insights related to the impact of LoS Rician and NLoS Rayleigh fading channels as well as the optimal ferrying distance are obtained considering a variety of objective functions.

Index Terms—UAV-assisted cellular communications, load-carry-and-deliver (LCAD), data ferrying, end-to-end outage probability, energy-efficiency, multi-objective optimization.

I. INTRODUCTION

UNMANNED aerial vehicle (UAV)-assisted data ferrying is a key enabling technology for wireless backhauling and multicasting in beyond 5G wireless networks. Different from conventional relaying, where a UAV establishes a communication link between the ground source and destination

simultaneously, data ferrying allows a UAV to load the data from a source, carry, and then deliver data to the destination when reaching sufficiently close to it [1]. This approach is known as *load carry and deliver (LCAD)* for fixed wing UAVs. The UAV-assisted data ferrying can be useful for ground receivers (either BSs or users) from various perspectives, such as (i) *Wireless backhauling*: Since the backhaul requirements can significantly vary depending on the locations of BSs (e.g. BSs in rural areas or hard-to-reach locations may not have well-established backhaul infrastructure [2]), wireless backhauling is an alternative to wired backhauling [3], [4]. Thus, UAV-assisted data ferrying provides a promising solution to assist wireless backhauling infrastructure [5], [6]. (ii) *Data dispatch in disaster situations*: UAV-assisted data ferrying becomes vital in disaster situations such as when the existing communication infrastructure including BSs has collapsed or where the communication infrastructure is not sufficient to fulfill the user requirements (i.e. most of the terrestrial BSs are congested). A similar idea referred to as “message ferrying” was studied in [7] for ad-hoc networks. The UAV data ferrying is perceived as a data dispatcher to multiple ground nodes in [8]. Data ferrying is beneficial for delay-tolerant applications where the source and the destination do not have a direct communication link.

However, the energy consumption of UAV-assisted data ferrying is crucial since the UAVs have limited on-board energy and the UAVs do not only consume communication energy but also significant propulsion energy to remain aloft or maintain their movement [9], [10]. The propulsion energy is directly proportional to the size of the UAV and the distance the UAV travels to ferry the data, whereas the communication energy depends on the channel fading and the path-loss factors etc. Accurate performance modeling and optimization of UAV-assisted data ferrying networks require precise energy consumption model, aerial path-loss and channel fading models.

A. Background Work

A series of research works [11]–[16] considered signal-to-noise ratio (SNR) outage characterization of UAV-assisted communications assuming either line-of-sight (LoS) Rician or non-LoS (NLoS) Nakagami- m faded aerial channels. That is, the impact of having a certain LoS and NLoS probability was ignored. Also, the derived expressions are generally in the form of complicated mathematical functions that cannot be

Manuscript received August 6, 2019; revised October 31, 2019; accepted December 1, 2019. Date of publication December 10, 2019; date of current version March 18, 2020. The associate editor coordinating the review of this article and approving it for publication was R. Zhang. (*Corresponding author: Ekram Hossain.*)

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Digital Object Identifier 10.1109/TCOMM.2019.2958805

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directly used for network planning and optimization purposes. For instance, [11], [12] provided closed-form expressions for the SNR outage probability assuming Nakagami- m fading aerial channels with no notion of LoS and NLoS transmissions. The bit error rate was also analyzed in [11] considering Nakagami- m fading channels. In addition, [13], [14] assumed Rician faded LoS aerial channels and derived the SNR outage in the form of Marcum Q -function. In [15], the SNR outage probability was analyzed for Rician and Rayleigh fading channels for LoS and NLoS channels, respectively. However, the final closed-form expressions are in the form of Marcum Q -function and Bessel function. The expressions derived in [11]–[15] are generally either in the form of special functions (such as Q -function or Hypergeometric function) or are represented using integrals. As such, their application to the optimization of network parameters is a fundamental challenge.

None of the aforementioned research works considered the performance characterization or optimization of UAV-assisted wireless data ferrying. The idea of UAV-assisted data ferrying (referred to as LCAD) was proposed in [1] assuming fixed wing UAV. Both the data loading and delivering links were considered. Similarly, [17] analyzed the throughput and delay trade-off of a similar data ferrying system where coverage to multiple ground nodes is provided by the UAV, i.e. the UAV delivers data to a node when reaching sufficiently close to it. Different from fixed-wing UAVs (where hovering is not possible), [8] proposed fly hover and communicate (FHC) protocol for rotary wing UAVs where multiple ground nodes are served with a single UAV. That is, the UAV flies to each user and communicate using optimal hovering location to minimize energy consumption and mission completion time. Successive iterative algorithms are solved at each location to find next optimal hovering location to serve each ground node. Note that the energy consumption of a rotary-wing UAV is fairly different from fixed-wing UAVs. In [18], the performance of a UAV was optimized when the UAV multicasts a data file to multiple ground receivers by designing the optimal trajectory that minimizes the mission time under a predefined quality of service (QoS) constraint. To minimize the mission completion time, the authors formulated a traveling salesman (TSM) problem for a UAV to communicate with each ground node individually and proposed random linear network coding (RLNC) to multicast the data [18]. Nonetheless, both [8], [17] considered only data transmission links and ignored the performance limitation caused by the data loading links.

B. Contributions

We consider the performance modeling and optimization of a UAV-assisted data ferrying network with Rician and Rayleigh fading for LoS and NLoS scenarios, respectively. Our contributions are as follows:

- We characterize the end-to-end performance of a UAV-assisted data ferrying considering (i) both the data loading and delivering links, (ii) both the LoS and NLoS transmissions in the data loading and delivering links, (iii) Rician fading model for LoS and Rayleigh fading

model for NLoS transmissions, and (iv) specific energy consumption model for the rotary wing UAV.

- We derive novel closed-form expressions for end-to-end network reliability¹ and bit-error probability (BEP) of the considered UAV-assisted data ferrying network. We define reliability as the packet error probability below a predefined reliability threshold. We characterize the reliability both with and without packet retransmission technique called automatic repeat request (ARQ). The derived expressions are novel compared to the expressions derived in conventional decode-and-forward (DF) relaying literature with Rician fading. Typically, they use single hop equivalent expression for BEP and do not consider averaging over the Rician fading channel [25].
- We derive end-to-end energy efficiency (a function of the SNR outage probability) and develop novel tractable approximations for the derived SNR outage. We demonstrate the use of these approximations in optimizing the distance that a UAV should travel to balance the energy-rate trade-offs.
- We formulate two different optimization problems and solve for the optimal ferrying distance, i.e. (i) outage-constrained energy minimization and (ii) energy-constrained SNR outage minimization. In the first problem, we convexify the problem and demonstrate that the obtained solution is 95% close to the solution from exhaustive search. To convexify the problem, we propose a modified approximation of the error function approximation presented in [26]. We provide a comparison to [26] and demonstrate that the proposed approximation is more accurate with polarity consistency. Closed-form optimal solutions are obtained for the second problem.
- We formulate a bi-objective optimization problem with the objectives to minimizing the SNR outage and energy consumption with desired SNR outage probability constraints. The objective function is then reformulated using difference of convex functions (DC) and solved using a DC algorithm.

Numerical results are presented to validate the derived expressions. Insights are extracted related to the impact of LoS Rician and NLoS Rayleigh fading channels on a variety of performance metrics as well as the optimal ferrying distance considering a variety of objective functions.

Although numerous research works [10], [27]–[31] have focused to date on energy efficiency and communication performance optimization, the considered problem of data ferrying has not been considered before in terms of performance characterization (i.e. reliability and coverage probability²). In addition to this, and most importantly, the proposed framework incorporates the derived expressions (after some transformations to tractable mathematical forms) into the considered

¹To date, reliability has been derived either based on the throughput [19], transmission outage probability [20], bit-error probability (BEP) or packet error probability (PEP) [21]–[24].

²BEP is appropriate to consider in scenarios vulnerable to small-scale fading e.g. mobile communications, whereas for large-scale fading such as shadowing and path-loss, outage probability is more relevant [32].

TABLE I
MATHEMATICAL NOTATIONS

Notation	Description	Notation	Description
γ_u, γ_k	UL SNR, DL SNR of k -th RX	Γ_0	End-to-end SNR threshold
p_u, p_d	UL and DL power consumption	P_h	power consumption in hovering state
R_0	Data rate requirement	$P(V)$	power consumption during flight
h_u, h_k	UL and DL small-scale fading gain	X_u, X_k	$ h_u ^2, h_k ^2$
θ_u, θ_k	UL and DL elevation angle	H	Height of UAV
d_k	Distance between UAV at $\mathbf{d}$ to k -th RX	$d; D$	Distance travelled by UAV; Distance between $\mathbf{I}$ & $\mathbf{F}$
$\mathcal{K}$	Number of RXs in downlink side	$K(\theta_u); K(\theta_k)$	Rician fading channel K-factor for UL and DL
$K_u; K_k$	Short representation of $K(\theta_u)$ & $K(\theta_k)$	d_0	Distance between UAV at distance d & $\mathbf{C}$
$\alpha(\theta_u); \alpha(\theta_k)$	Path-loss exponent for UL & DL	$\alpha_u; \alpha_k$	short representation of $\alpha(\theta_u)$ & $\alpha(\theta_k)$
k	RX index at downlink	N_0	Noise power spectral density
R	Downlink cell radius	T_u	Hovering time in UL
T_d	Hovering time of UAV in downlink	T_c	Transmission time of UAV
$p_L(\theta_u); p_N(\theta_u)$	Probability of LoS & NLoS in UL	$p_L(\theta_k); p_N(\theta_k)$	Probability of LoS & NLoS of k -th RX in DL
v_i	Event based on UL & DL LoS probability	L	Packet length
$P_b^k(v_i, d_k)$	End-to-end conditional bit error probability	$\mathbb{P}_\epsilon^k(v_i, d_k)$	End-to-end packet error probability
$P_b^{u,L}; P_b^{u,N}$	Bit error probability in UL LoS & UL NLoS	$P_\epsilon^{k,L}; P_\epsilon^{k,N}$	PEP DL LoS & DL NLoS for BS_k conditioned on d_k & v_i
$P_b^k(d_k)$	End-to-end bit error probability conditioned on d_k only	(r_k, ϕ_k)	Coordinates of BS_k w.r.t., cell centre $\mathbf{C}$
τ_R, τ	Threshold on reliability; threshold on outage probability	$\Omega_u; \Omega_k$	Mean local power of UL and DL of BS_k
$\mathbb{P}(v_i)$	Probability of occurrence of event v_i	$F_{X_u}^L; F_{X_u}^N$	UL CDF for LoS and NLoS

optimization problems, which is different from the typical computationally intensive instantaneous optimization frameworks where the objective function and constraints are defined for each fading channel realization (which is typically assumed to be perfectly known).

The remaining of the paper is organized as follows. We describe the system model in Section II. In Section III, we characterize the reliability and BEP for the considered network. In Section IV, we derive the end-to-end energy efficiency and the SNR outage probability for the considered network. The approximations for the Rician fading channels are also presented in Section IV. The optimization problems are formulated in Section V. Then, we present the numerical results in Section VI before the conclusions in Section VII. A list of important variables is in **Table I**.

II. SYSTEM MODEL AND ASSUMPTIONS

A. UAV-Assisted Data Ferrying Model

We consider a UAV-assisted data ferrying network in which a rotary-wing UAV is responsible to ferry and transmit the data taken from a source BS BS_s to $\mathcal{K}$ destined RXs. The communication is established from BS_s located at $\mathbf{S}(x_s, y_s, 0)$ to UAV and then UAV to k -th ground receiver RX_k located at $(x_k, y_k, 0)$ or (r_k, ϕ_k) . There is no direct connection available between the source BS and destination RXs. Each RX_k is uniformly distributed in a circular region of radius R around the cell center $\mathbf{C}(x_c, y_c, 0)$. Since the destination RXs are uniformly distributed, the distribution of their respective distances from $\mathbf{C}$ is given as $f_r(r_k) = \frac{2r_k}{R^2}$, $0 \leq r_k \leq R$.

The UAV loads the data from the source BS_s which is at distance $d_u = H$ from the UAV and carries it to a distance d from $\mathbf{I}$ prior to the downlink transmission. The initial and final locations of the UAV flight, i.e. $\mathbf{I}(x_s, y_s, H)$ and $\mathbf{F}(x_c, y_c, H)$ are located right above $\mathbf{S}$ and $\mathbf{C}$, respectively. Here, $d = 0$ corresponds to the situation when the UAV decides to transmit from $\mathbf{I}$ and minimizes its energy

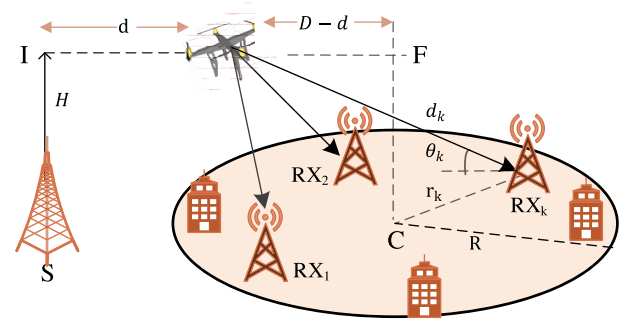


Fig. 1. UAV-assisted data ferrying for multicasting and serving K terrestrial receivers.

consumption due to flight and $d = D$ corresponds to the situation when UAV decides to transmit from $\mathbf{F}$ and minimizes the communication outage probability. Note that, $0 \leq d \leq D$ and $D = \sqrt{(x_s - x_c)^2 + (y_s - y_c)^2}$.

The uplink (data loading) and downlink (data delivering) transmissions use the same frequency channel but occur at different time slots. In the downlink, the UAV transmits the same data to each of the $\mathcal{K}$ RXs simultaneously (i.e. we consider a **multicasting** scenario). Each of the $\mathcal{K}$ RXs has a fixed data rate requirement R_0 defined as $R_0 = B \log_2(1 + \Gamma_0)$, where Γ_0 represents the minimum end-to-end SNR threshold to achieve R_0 , i.e. $\Gamma_0 = 2^{\frac{R_0}{B}} - 1$. Here, B represents the transmission bandwidth. Note that the UAV can travel up to a maximum distance D from $\mathbf{I}$, however, the UAV may stop in between $\mathbf{I}$ and $\mathbf{F}$ when the desired data rate is met (Fig. 1).

B. Aerial Channel Model

Depending on the elevation angle between the UAV and ground nodes (or altitude of the UAV) and the environment (e.g. the intensity and heights of buildings), the transmission

to the ground users may have LoS or non-LoS channel. The elevation angle (in rad) between the UAV and BS_k is given as follows:

$$\begin{aligned}\theta_k &= \arctan\left(\frac{H}{\sqrt{(x_k - d)^2 + y_k^2}}\right) \\ &= \arctan\left(\frac{H}{\sqrt{(r_k \cos(\phi_k) - d)^2 + r_k^2 \sin(\phi_k)^2}}\right).\end{aligned}\quad (1)$$

Similarly, the elevation angle between the UAV and BS_s is $\theta_u = \pi/2$ rad. The probability of LoS in the uplink and downlink is a function of θ_u and θ_k , respectively, i.e.

$$p_L(\theta_u) = (e_u \exp(-g_u(\theta_u - e_u)) + 1)^{-1}, \quad (2)$$

$$p_L(\theta_k) = (e_k \exp(-g_k(\theta_k - e_k)) + 1)^{-1}, \quad (3)$$

where e_u , e_k , g_u , and g_k are the environment parameters obtained from the curve fitting using Damped Least-Squares (DLS) method [33]. We model the LoS and NLoS transmissions with Rician and Rayleigh fast fading channels, respectively. For LoS Rician fading, the probability density function (PDF) of the fading channel power X_L is given as:

$$\begin{aligned}f_{X_L}(x) &= \frac{K+1}{\Omega} e^{-K - \frac{(K+1)x}{\Omega}} I_0\left(2\sqrt{\frac{K(K+1)x}{\Omega}}\right) \\ &= \sum_{\ell=0}^{\infty} \frac{b(bK)^{\ell}}{(\ell!)^2} x^{\ell} e^{-bx-K},\end{aligned}\quad (4)$$

in which K is the Rician factor and I_0 is a modified Bessel function of the first kind and $b = \frac{K+1}{\Omega}$ [34]. The PDF of NLoS Rayleigh fading channel power is obtained by substituting $K = 0$ in (4) as follows:

$$f_{X_N}(x) = \frac{1}{\Omega} \exp\left(-\frac{x}{\Omega}\right), \quad (5)$$

where Ω is the mean local power of the fading channel. The path-loss exponent α and K -factor for the Rician uplink channel are a function of the elevation angle, i.e. [35]

$$\alpha(\theta_u) = p_L(\theta_u)q_u + v_u, \quad K(\theta_u) = w_u \exp(z_u \theta_u). \quad (6)$$

Here q_u , v_u , w_u , and z_u are constants depending on the uplink environment [33]. Similar to (6), for the downlink we can write

$$\alpha(\theta_k) = p_L(\theta_k)q_d + v_d, \quad K(\theta_k) = w_d \exp(z_d \theta_k). \quad (7)$$

respectively. Again, q_d, v_d, w_d , and z_d are constants depending on the downlink environment [33].

C. Uplink and Downlink Transmission Model

The uplink received signal at the UAV can be defined as $y_u = \sqrt{\hat{A}p_u\eta_u^{-1}d_u^{-\alpha(\theta_u)}} h_u s + w_u$, where s is the binary phase shift keying (BPSK) signal transmitted from BS_s to the UAV, η_u denotes the excess aerial path-loss, p_u is the uplink transmission power of BS_s, d_u is the distance between the BS_s and the UAV located at **I**, i.e. $d_u = H$ (Fig. 1), $\hat{A}$ reflects system parameters (e.g. operating frequency and antenna gain), w_u is the noise which follows zero-mean Gaussian distribution

with power spectral density N_0 , and h_u is the channel fading gain. The uplink signal-to-noise-ratio (SNR) is formulated as:

$$\gamma_u = \frac{\hat{A}p_u\eta_u^{-1}H^{-\alpha(\theta_u)}X_u}{N_0}, \quad (8)$$

where $X_u = |h_u|^2$. The distribution of X_u follows from a non-central chi-square distribution given in (4) for LoS and exponential distribution given in (5) for NLoS.

In the downlink, the UAV communicates to $\mathcal{K}$ RXs. The distance between the UAV at d and the k -th ground RX is represented by $d_k, \forall k$ (Fig. 1). The received signal y_k at k -th RX is

$$y_k = \sqrt{\hat{A}p_d\eta_d^{-1}d_k^{-\alpha(\theta_k)}} h_k y_u + w_k,$$

where η_d and p_d denote the excess aerial path-loss and downlink transmission power of UAV, respectively. The SNR received at k -th RX in the downlink can then be expressed as:

$$\gamma_k = \frac{\hat{A}p_d\eta_d^{-1}d_k^{-\alpha(\theta_k)}X_k}{N_0}, \quad (9)$$

where $X_k = |h_k|^2$. The distribution of X_k follows from a non-central chi-square distribution and exponential distribution given in (4) for LoS and (5) for NLoS, respectively. The distance between the UAV and RX k is given as $d_k = \sqrt{(D - d - r_k \cos(\phi_k))^2 + (r_k \sin(\phi_k))^2 + H^2}$. **Note that from this point onward we will use K_u as a short notation for $K(\theta_u)$, K_k as a short notation for $K(\theta_k)$, α_u and α_k as $\alpha(\theta_u)$ and $\alpha(\theta_k)$ for brevity, respectively.**

D. Energy Consumption Model

The total energy consumption of a UAV comprises of (i) propulsion energy which is the energy consumed by UAV in traveling from one location to the other location (plus hovering energy in case of rotary wing UAVs), and (ii) transmission energy is the energy consumed by UAV for communication purposes. The propulsion energy (typically in the order of hundreds of watts [36]) is required to keep the UAV aloft and fly forward/vertically. The communication energy of UAV is in the order of tens of watts [37]. A trade-off exists between the communication and propulsion energy as a function of the distance that UAV travels while ferrying the data. The consumed energy for the data loading phase, UAV travelling phase, and downlink transmission phase are given in the following.

1) *Uplink Data Loading Phase:* The UAV hovers at **I** for a predefined duration³ T_u (Fig. 1). The energy consumption due to hovering of UAV and communication of the source BS in the uplink [8] is given as:

$$E^u = (P_h + p_u)T_u, \quad (10)$$

where p_u is the uplink power consumption of BS_s and P_h is the propulsion power consumption of UAV in the hovering

³We assume that transmission time T_u is the time data takes to reach to the UAV which is at location **I** (e.g. $T_u = \frac{H}{c}$). To assure that complete data is uploaded to UAV from BS_s, we can add an extra delay to the transmission time.

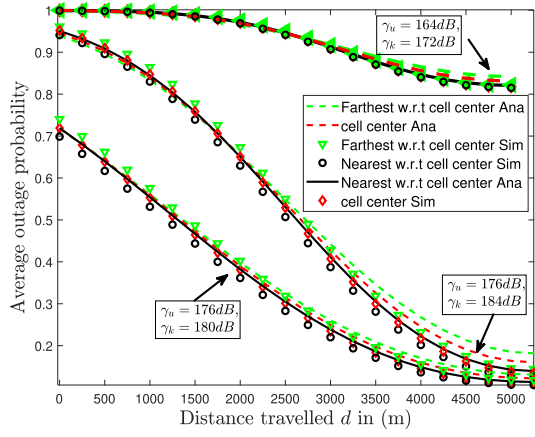


Fig. 2. Outage of receivers that are located nearest and farthest from the cell-center as well as the receiver located at the cell-center, considering Rayleigh/exponential distribution $\alpha = 2.5$ as a function of d .

state given as follows:

$$P_h = P_0 + P_i = \frac{\delta}{8} \rho s A \xi^3 r^3 + (1 + \kappa) \sqrt{\frac{(mg)^3}{2\rho A}}, \quad (11)$$

where ρ , A , ξ , r , s , δ , and κ denote the air density (in kg/m³), rotor disc area (in m²), blade angular velocity (in rad/sec), rotor radius (in m), rotor solidity, profile drag coefficient, and incremental correction factor of induced power, respectively. The total weight W of the UAV is approximately equal to the gravitational force, i.e. $W = mg$ (in Newton), where m is the UAV mass (in kg) and g denotes earth gravity (in m/s²).

2) *UAV Travelling Phase*: The energy consumption during the travelling time is solely due to the propulsion energy as the UAV is not communicating during the flight. When the UAV moves with the constant velocity V and carries the data to a distance d before delivering to downlink BS_s, the energy consumption can be modeled as follows [8]:

$$E^t(d) = P(V)d/V, \quad (12)$$

where $P(V)$ is the power consumed by the blade profile, induced power, and parasitic power of the UAV, i.e.

$$P(V) = P_0 \left(1 + \frac{3V^2}{U_{\text{tip}}^2} \right) + P_i \left(\sqrt{1 + \frac{V^4}{4v_0^4}} - \frac{V^2}{2v_0^2} \right) + \frac{1}{2} \hat{d}_0 \rho s A V^3,$$

where U_{tip} denotes the tip speed of the rotor blade, v_0 is the mean rotor induced velocity, and $\hat{d}_0$ is the fuselage drag ratio. By substituting $V = 0$ in the aforementioned equation, we get the hovering power as given in (11). Note that $E^t(d)$ is directly proportional to the distance traveled by the UAV d .

3) *Downlink Transmission Phase*: In the downlink, the UAV experiences energy consumption due to communication and hovering. We consider that the serving region (i.e. the cell radius R) in which all receivers are located is very small compared to the traveling distance of UAV $r_k = 0$, i.e. $d_k = d_0 = \sqrt{(D-d)^2 + H^2}$ for simplicity. Under this assumption, the farthest or nearest receiver becomes approximately the same as the receiver who is located at the cell-center (see Figure 2).

As such, we consider the outage probability, with respect to the cell-center, as an approximate performance measure of all users in the serving region. The UAV hovering time T_d is assumed to be at least equal to the time in which a user located at the cell centre receives data from UAV, where T_c is the transmission time of the UAV. It is important to note that the UAV can transmit at any specific distance d , from the initial point on the trajectory, if the QoS criterion is met. Therefore, the traveling time of UAV to location d is separate from the transmission time in the downlink T_c . The downlink energy consumption at UAV can thus be modeled as follows:

$$E^d(d) = P_h T_d + p_d T_c = P_h \frac{d_0}{c} + p_d T_c, \quad (13)$$

where $d_0 = \sqrt{(D-d)^2 + H^2}$, c is the speed of light, and T_d is the hovering time of UAV. Since the UAV will be transmitting for the entire hovering duration, we assume that $T_d = T_c$.

4) *End-to-End Energy Consumption*: The overall energy consumption can be obtained by adding (10), (12), and (13) as follows:

$$E_{\text{tot}}(d) = \frac{(P_h + p_d) \sqrt{(D-d)^2 + H^2}}{c} + \frac{P(V)d}{V} + C, \quad (14)$$

where $C = p_u T_u + P_h T_u$. We assume that sufficiently large memory is available at the UAV to store the complete data for one trip.

III. END-TO-END RELIABILITY AND BIT ERROR PROBABILITY ANALYSIS

In this section, we derive the end-to-end reliability with and without ARQ and BEP (the ratio of the number of bits in error to the total number of transmitted bits) of the considered network. The end-to-end BEP and PEP for the k -th RX are conditioned on the following events, i.e.

(i) **Event v_1** : when both the uplink and downlink incur LoS Rician fading, i.e. $\{p_L(\theta_u) > p_N(\theta_u), p_L(\theta_k) > p_N(\theta_k)\}$, where $p_N(\cdot)$ depicts the NLoS probability,

(ii) **Event v_2** : when the uplink experiences LoS Rician fading and the downlink experiences NLoS Rayleigh fading, i.e. $\{p_L(\theta_u) > p_N(\theta_u), p_L(\theta_k) < p_N(\theta_k)\}$,

(iii) **Event v_3** : when the uplink experiences NLoS Rayleigh fading and the downlink experiences LoS Rician fading, i.e. $\{p_L(\theta_u) < p_N(\theta_u), p_L(\theta_k) > p_N(\theta_k)\}$, and

(iv) **Event v_4** : when the uplink and downlink experience NLoS and NLoS Rayleigh fading, i.e. $\{p_L(\theta_u) < p_N(\theta_u), p_L(\theta_k) < p_N(\theta_k)\}$.

Conditioned on v_i , d_k and using the definition of reliability from [22], the end-to-end PEP of k -th RX $\mathbb{P}_e^k(v_i, d_k)$ should meet the reliability threshold (or packet error threshold) τ_R [22]. We model the reliability constraint without ARQ as:

$$\mathbb{P}_e^k(v_i, d_k) \leq \tau_R, \quad (15)$$

where a packet is assumed to be in error even if a single bit is in error and we assumed that the bit errors are independent

(e.g. due to large interleaver size, fast fading and uncoded transmission); therefore, we have

$$\mathbb{P}_\epsilon^k(v_i, d_k) = 1 - (1 - P_b^k(v_i, d_k))^L, \quad (16)$$

where L is the number of bits per packet with the assumption that cyclic redundancy check (CRC) is not available at the UAV in uplink, and $P_b^k(v_i, d_k)$ is the conditional BEP given as $P_b^k(v_i, d_k) \leq 1 - (1 - \tau_R)^{\frac{1}{L}}$.

Remark: The end-to-end PEP with ARQ and n retransmissions $\mathbb{P}_\epsilon^k(n, v_i, d_k)$ can be obtained from the BEP under the assumption of i.i.d. channels [38] in each retransmission (e.g. due to large interleaver size, fast fading and uncoded retransmissions); for both uplink and downlink as

$$\mathbb{P}_\epsilon^k(n, v_i, d_k) = 1 - (1 - P_\epsilon^u(n))(1 - P_\epsilon^k(n)), \quad (17)$$

where $n = 0, 1, \dots, \mu$ and μ is the maximum number of retransmissions. The PEP of both uplink and downlink are $P_\epsilon^u(n) = (1 - (1 - P_b^u)^L)^{n+1}$ and $P_\epsilon^k(n) = (1 - (1 - P_b^k)^L)^{n+1}$, respectively. It is assumed that both UAV and downlink receivers have single bit feedback associated to it. The reliability constraint in (15) can be rewritten by considering ARQ as $\mathbb{P}_\epsilon^k(n, v_i, d_k) \leq \tau_R$. For $n = 0$ and no CRC in the uplink (i.e. $P_\epsilon^u(n) = 0$, $P_\epsilon^k(n) = \mathbb{P}_\epsilon^k(v_i, d_k)$), $\mathbb{P}_\epsilon^k(n, v_i, d_k)$ converges to (16).

A. Methodology of Analysis

To derive the reliability and $\mathbb{P}_b^k(v_i, d_k)$, the steps are as follows:

- 1) Derive the uplink BEP $P_b^{u,L}$ and $P_b^{u,N}$ for LoS (Rician) and NLoS (Rayleigh) fading channels, respectively. Derive the downlink BEP $P_b^{k,L}$ and $P_b^{k,N}$ for LoS (Rician) and NLoS (Rayleigh) fading for BS_k , respectively.
- 2) The bit error can occur in two scenarios [39]: (i) when the uplink transmission is in error but the downlink transmission is successful (ii) when the uplink transmission is successful but the downlink transmission is in error. Conditioned on d_k and v_i , the end-to-end BEP of k -th RX is [25], [39]:

$$P_b^k(v_i, d_k) = (1 - P_b^u) P_b^k + P_b^u (1 - P_b^k), \quad (18)$$

where P_b^u and P_b^k are the uplink and downlink conditional BEPs for k -th link, respectively, given v_i and d_k . For events v_1, v_2, v_3 , and v_4 , (18) can be rewritten as:

$$\begin{aligned} P_b^k(v_1, d_k) &= (1 - P_b^{u,L}) P_b^{k,L} + P_b^{u,L} (1 - P_b^{k,L}), \\ P_b^k(v_2, d_k) &= (1 - P_b^{u,L}) P_b^{k,N} + P_b^{u,L} (1 - P_b^{k,N}), \\ P_b^k(v_3, d_k) &= (1 - P_b^{u,N}) P_b^{k,L} + P_b^{u,N} (1 - P_b^{k,L}), \\ P_b^k(v_4, d_k) &= (1 - P_b^{u,N}) P_b^{k,N} + P_b^{u,N} (1 - P_b^{k,N}). \end{aligned} \quad (19)$$

- 3) Derive the BEP of multicasting system ($P_b^{MC}(d_k)$). The unconditioned BEP (after averaging over v_i and d_k) for any arbitrary RX can then be given as follows:

$$P_b = \sum_{i=1}^4 \mathbb{P}(v_i) \mathbb{E}_{d_k} [P_b^k(v_i, d_k)], \quad (20)$$

where $\mathbb{P}(v_i)$ is the probability of event v_i .

B. Characterization of Uplink BEP - ($P_b^{u,L}$ and $P_b^{u,N}$)

Proposition 1 BEP $P_b^{u,L}$ - LoS Rician Fading: The BEP in the uplink LoS transmission channel (i.e. Rician fading) can be derived as follows:

$$P_b^{u,L} = \sum_{\ell=0}^{\infty} q_u(\ell) (\beta_1^{-\ell-1} + 3\beta_2^{-\ell-1}), \quad (21)$$

where $q_u(\ell) = \frac{b_u e^{-K_u} (b_u K_u)^\ell}{12\ell!}$, $\beta_1 = \frac{\hat{A} p_u \eta_u^{-1}}{2H^{\alpha_u} N_0} + b_u$, $\beta_2 = \frac{2\hat{A} p_u \eta_u^{-1}}{3H^{\alpha_u} N_0} + b_u$, and $b_u = \frac{K_u + 1}{\Omega_u}$.

Proof: By conditioning on $X_u = |h_u|^2$, the BEP in the uplink can be written using the BEP of Gaussian noise as:

$$P_b^{u,L} = \mathbb{E}_{X_u} \left[Q \left(\sqrt{2(\beta_1 - b_u) X_u} \right) \right]. \quad (22)$$

Using the exponential approximation for Q-function [40], i.e. $Q(x) = \frac{1}{12} e^{-\frac{x^2}{2}} + \frac{1}{4} e^{-\frac{2x^2}{3}}$, we simplify (22) as follows:

$$P_b^{u,L} = \mathbb{E}_{X_u} \left[\frac{1}{12} e^{-(\beta_1 - b_u) X_u} + \frac{1}{4} e^{-(\beta_2 - b_u) X_u} \right]. \quad (23)$$

The first expectation over X_u in (23) can be solved by averaging over the PDF of non-central chi-square distribution given in (4) and by substituting z with X_u and b with b_u . The first term of (23) can be solved as follows:

$$\frac{1}{12} \mathbb{E}_{X_u} [e^{-(\beta_1 - b_u) X_u}] = \sum_{\ell=0}^{\infty} q_u(\ell) \beta_1^{-\ell-1}, \quad (24)$$

where the closed-form is calculated using the definition of Gamma function in (24), i.e. $\int_0^\infty e^{-\beta_1 x} x^\ell dx = \beta_1^{-\ell-1} \Gamma(\ell + 1)$. Following the same steps, we solve the second expectation in (23) as:

$$\mathbb{E}_{X_u} \left[\frac{1}{4} e^{-(\beta_2 - b_u) X_u} \right] = \sum_{\ell=0}^{\infty} q_u(\ell) \beta_2^{-\ell-1}. \quad (25)$$

Substituting (24) and (25) in (23), we get **Proposition 1**. ■

Corollary 1 BEP $P_b^{u,N}$ - NLoS Rayleigh Fading: Substituting $K_u = 0$ in Proposition 1, the BEP in the NLoS Rayleigh fading channel (21) is given as follows:

$$P_b^{u,N} = \frac{1}{12\Omega_u} (\beta_1^{-1} + 3\beta_2^{-1}), \quad (26)$$

where β_1 and β_2 are given as in **Proposition 1**.

C. Characterization of Downlink BEP - $P_b^{k,L}$ and $P_b^{k,N}$

Proposition 2 BEP $P_b^{k,L}$ - LoS Rician Fading: The BEP P_b^k in the downlink for k -th RX can be derived as follows:

$$P_b^{k,L} = \sum_{\ell=0}^{\infty} q_d(\ell) \left(\left(\frac{\beta'_1}{d_k^{\alpha_k}} + b_k \right)^{-\ell-1} + 3 \left(\frac{\beta'_2}{d_k^{\alpha_k}} + b_k \right)^{-\ell-1} \right), \quad (27)$$

where $q_d(\ell) = \frac{b_d^{\ell+1} e^{-K_k} K_k^\ell}{12(\ell!)}$, $\beta'_1 = \frac{\hat{A} p_d \eta_k^{-1}}{2N_0}$, $\beta'_2 = \frac{2\hat{A} p_d \eta_k^{-1}}{3N_0}$ and $b_k = \frac{K_k + 1}{\Omega_d}$.

Proof: The proof can be done following similar steps as in the proof of **Proposition 1**. ■

Corollary 2 $BEP P_b^{k,N}$ - NLoS Rayleigh Fading: The $BEP P_b^{k,N}$ in the downlink NLoS transmission link can be given by substituting $K_k = 0$ in (27) as follows:

$$P_b^{k,N} = \frac{1}{12\Omega_d} \left(\left(\frac{\beta'_1}{d_k^{\alpha_k}} + \frac{1}{\Omega_d} \right)^{-1} + 3 \left(\frac{\beta'_2}{d_k^{\alpha_k}} + \frac{1}{\Omega_d} \right)^{-1} \right), \quad (28)$$

where β'_1 and β'_2 are given as in **Proposition 2**.

D. End-to-End BEP of RX k , $P_b^k(d_k)$

Conditioned on d_k , the end-to-end BEP $P_b^k(d_k)$ can be derived as follows:

$$P_b^k(d_k) = \sum_{i=1}^4 \mathbb{P}(v_i) P_b^k(v_i, d_k), \quad (29)$$

where $\mathbb{P}(v_i)$ represents the probability of event v_i . The occurrence of each event can be written as the product of two individual events, i.e. $\mathbb{P}(v_1) = p_L(\theta_u)p_L(\theta_k)$, $\mathbb{P}(v_2) = p_L(\theta_u)p_N(\theta_k)$, $\mathbb{P}(v_3) = p_N(\theta_u)p_L(\theta_k)$, $\mathbb{P}(v_4) = p_N(\theta_u)p_N(\theta_k)$. Note that $p_N(\cdot) = 1 - p_L(\cdot)$.

Also, $P_b^k(v_1, d_k)$ is given by using (21) and (27) in (18). $P_b^k(v_2, d_k)$ is given by using (21) and (28) in (18). $P_b^k(v_3, d_k)$ is given by substituting (26) and (27) in (18). $P_b^k(v_4, d_k)$ is obtained by using (26) and (28) in (18).

E. Overall BEP of the MultiCasting System, $P_b^{MC}(d_k)$

Conditioned on d_k , the overall BEP of the considered multicasting network can be expressed as follows:

$$P_b^{MC}(d_k) = \sum_{k=1}^{2^{\mathcal{K}}-1} \frac{1}{2^{\mathcal{K}}-1} \prod_{k=1}^{\mathcal{K}} (1 - P_b^k(d_k))^{1-s_k} (P_b^k(d_k))^{s_k}. \quad (30)$$

To elaborate the above expression, let us consider a simple example of two RXs $\mathcal{K} = 2$. Each RX $k \in \mathcal{K}$ can experience successful or erroneous transmission denoted by $s_k = 0$ and $s_k = 1$, respectively. $\mathbf{S}$ represents the set containing all possible combinations (i.e. $2^{\mathcal{K}}$ combinations). For two RXs, we have $\mathbf{S} = \{00, 01, 10, 11\}$ where each combination is $\mathbf{s} \in \mathbf{S}$. The first combination $\mathbf{s} = \{00\}$ corresponds to both RXs having no errors so need not be considered in the overall BEP. Then, $\mathbf{s} = \{01\}$ corresponds to the event when first RX in success while the second RX is in error with BEP given by $(1 - P_b^1(d_1))P_b^2(d_2)$. Similarly for $\mathbf{s} = \{10\}$ and $\mathbf{s} = \{11\}$, the corresponding BEPs are $P_b^1(d_1)(1 - P_b^2(d_2))$ and $P_b^1(d_1)P_b^2(d_2)$, respectively. Averaging over all combinations with at least one error, we obtain (30).

On the other hand, when s_k is known *a priori* for each RX k based on certain QoS constraints, the overall BEP can be expressed as follows:

$$P_b^{MC}(d_k) = \prod_{k=1}^{\mathcal{K}} (P_b^k(d_k))^{s_k} (1 - P_b^k(d_k))^{1-s_k}, \quad (31)$$

where

$$s_k = \begin{cases} 1 & P_b^k(v_i, d_k) > 1 - (1 - \tau_R)^{1/L} \\ 0 & P_b^k(v_i, d_k) \leq 1 - (1 - \tau_R)^{1/L}. \end{cases} \quad (32)$$

That is, s_k obtained by solving (15) and (16).

F. BEP of Any Arbitrary BS

Finally, deconditioning over d_k (which in turn implies averaging over r_k and ϕ_k) yields the BEP P_b^k as:

$$\begin{aligned} P_b &= 1 - \int_0^R \int_0^{2\pi} P_b^k(d_k) \frac{r_k}{\pi R^2} dr_k d\phi_k \\ &= 1 - \sum_{i=1}^4 \int_0^R \int_0^{2\pi} \mathbb{P}(v_i) P_b^k(v_i, d_k) \frac{r_k}{\pi R^2} dr_k d\phi_k. \end{aligned} \quad (33)$$

The integrals can be solved using standard mathematical software such as MATLAB and MATHEMATICA.

It is important to note that our analysis can be extended for the nearest (farthest) user by replacing the probability density function (PDF) of the distance of the k -th user from cell-centre with the PDF of the user who is located at the minimum (maximum) distance from the cell-center using order statistics.

IV. END-TO-END ENERGY-EFFICIENCY AND SNR OUTAGE PROBABILITY

In this section, we characterize the network energy-efficiency and SNR outage probability of the considered multicasting network. Energy-efficiency (EE) is defined as the ratio of the number of successfully delivered bits and the total network energy consumption (in bits/Joule) [41]. The analytical methodology to derive EE is as follows:

- 1) Conditioned on v_i and d_k , characterize the end-to-end EE of k -th RX as follows:

$$EE(v_i, d_k) = \frac{R_0}{BE_{tot}(d)} (1 - O_k(d_k)), \quad (34)$$

where R_0 and B are the desired rate and channel bandwidth, respectively. Energy consumption $E_{tot}(d)$ is given in (14) and $O_k(d_k)$ is the SNR outage of k -th BS.

- 2) Characterize the SNR outage $O_k(d_k)$ of the k -th BS.
- 3) Characterize the overall SNR outage $O_k^{MC}(d_k)$ of the multicasting system.
- 4) Characterize the unconditioned end-to-end outage probability of any arbitrary RX as follows:

$$O = \sum_{i=1}^4 \mathbb{E}_{d_k} [\mathbb{P}(v_i) O_k(v_i, d_k)]. \quad (35)$$

- 5) Characterize the overall EE^{MC} of the network.

A. End-to-End Outage Probability of RX k , $O_k(d_k)$

Assuming that the UAV can perform decoding and storing of the data during flight, the end-to-end SNR Γ_k for k -th RX can be modeled as [42]:

$$\Gamma_k = \min\{\gamma_u, \gamma_k\}. \quad (36)$$

Conditioned on d_k and v_i , the end-to-end transmission outage probability of k -th RX is defined as:

$$\begin{aligned} O_k(v_i, d_k) &= \mathbb{P}(\Gamma_k < \Gamma_0 | v_i) = \mathbb{P}[\min(\gamma_u, \gamma_k) < \Gamma_0 | v_i] \\ &= 1 - (1 - F_{\gamma_u}(\Gamma_0)) (1 - F_{\gamma_k}(\Gamma_0)), \end{aligned} \quad (37)$$

where $F_{\gamma_u}(\Gamma_0)$ and $F_{\gamma_k}(\Gamma_0)$ represent the CDF of uplink and downlink SNR evaluated at Γ_0 (i.e. uplink and downlink conditional SNR outage), respectively, for the k -th BS.

For events v_1, v_2, v_3 , and v_4 , (37) can be rewritten as:

$$\begin{aligned} O_k(v_1, d_k) &= 1 - (1 - F_{\gamma_u}^L(\Gamma_0)) (1 - F_{\gamma_k}^L(\Gamma_0)), \\ O_k(v_2, d_k) &= 1 - (1 - F_{\gamma_u}^L(\Gamma_0)) (1 - F_{\gamma_k}^N(\Gamma_0)), \\ O_k(v_3, d_k) &= 1 - (1 - F_{\gamma_u}^N(\Gamma_0)) (1 - F_{\gamma_k}^L(\Gamma_0)), \\ O_k(v_4, d_k) &= 1 - (1 - F_{\gamma_u}^N(\Gamma_0)) (1 - F_{\gamma_k}^N(\Gamma_0)). \end{aligned} \quad (38)$$

To evaluate (38), we characterize $F_{\gamma_u}^L(\Gamma_0)$, $F_{\gamma_u}^N(\Gamma_0)$, $F_{\gamma_k}^L(\Gamma_0)$ and $F_{\gamma_k}^N(\Gamma_0)$, respectively, in the following.

1) *Characterization of $F_{\gamma_u}^L(\Gamma_0)$ and $F_{\gamma_u}^N(\Gamma_0)$* : Using (8), the uplink SNR outage can be given as follows:

$$F_{\gamma_u}(\Gamma_0) = \mathbb{P}(\gamma_u \leq \Gamma_0) = \mathbb{P}(X_u \leq \Gamma'_u) = F_{X_u}(\Gamma'_u), \quad (39)$$

where $X_u = |h_u|^2$ represents non-central chi square distribution and $\Gamma'_u = \frac{N_0 \Gamma_0}{\hat{A} p_u \eta_u^{-1} H^{-\alpha_u}}$. Using the alternate exact expression for PDF in (4), the CDF of non-central chi-square distribution can be given as follows [34], [43], [44]:

$$F_X(x) = 1 - \sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} f(m, \ell) x^m e^{-bx}, \quad (40)$$

where $f(m, \ell) = e^{-K} \frac{K^\ell b^m}{\ell! m!}$, $b = \frac{K+1}{\Omega}$, Ω is the mean local power of the Rician channel, and K is the Rician factor.

After substituting $x = \Gamma'_u$ in (40), $F_{\gamma_u}^L(\Gamma_0) = F_{X_u}^L(\Gamma'_u)$ for LoS fading can be evaluated as follows:

$$F_{X_u}^L(\Gamma'_u) = 1 - \sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} f_u(m, \ell) (\Gamma'_u)^m \exp(-b_u \Gamma'_u), \quad (41)$$

where $b_u = \frac{K_u+1}{\Omega_u}$, $f_u(m, \ell) = e^{-K_u} \frac{K_u^\ell b_u^m}{\ell! m!}$, and Ω_u is the mean local power of the Rician channel in uplink.

Substituting $K_u = 0$ in (41), $F_{X_u}^N(\Gamma'_u)$ for NLoS Rayleigh fading can be expressed as follows:

$$F_{X_u}^N(\Gamma'_u) = 1 - e^{-\Gamma'_u/\Omega_u}, \quad (42)$$

where Γ'_u is defined as in (39).

2) *Characterization of $F_{\gamma_k}^L(\Gamma_0)$ and $F_{\gamma_k}^N(\Gamma_0)$* : Using (9), the downlink SNR outage of k -th RX can be given as follows:

$$\begin{aligned} F_{\gamma_k}^L(\Gamma_0 | d_k) &= \mathbb{P}\left(\frac{\hat{A} p_d \eta_d^{-1} d_k^{-\alpha_k} X_k}{N_0} \leq \Gamma_0\right) = F_{X_k}^L(\Gamma'_k | d_k) \\ &\stackrel{(a)}{=} 1 - \sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} f_d(m, \ell) (\Gamma'_k d_k^{\alpha_k})^m \exp(-b_k \Gamma'_k d_k^{\alpha_k}), \end{aligned} \quad (43)$$

where $d_k = \sqrt{(D - d - r_k \cos(\phi_k))^2 + (r_k \sin(\phi_k))^2 + H^2}$, $\Gamma'_k = \frac{N_0 \Gamma_0}{\hat{A} p_d \eta_d^{-1}}$, and (a) follows by taking $x = \Gamma'_k d_k^{\alpha_k}$ in (40).

Also, $b_k = \frac{K_k+1}{\Omega_d}$, $f_d(m, \ell) = e^{-K_k} \frac{K_k^\ell b_k^m}{\ell! m!}$, and Ω_d is the mean local power of the downlink Rician channel.

Substituting $K_k = 0$ in (43), $F_{X_k}(\Gamma'_k)$ for NLoS Rayleigh fading can be expressed as follows:

$$F_{X_k}^N(\Gamma'_k | d_k) = 1 - e^{-\frac{\Gamma'_k}{\Omega_d} d_k^{\alpha_k}}. \quad (44)$$

3) *End-to-End SNR Outage, $O_k(d_k)$* : Conditioned on d_k , we write $O_k(d_k)$ as follows:

$$O_k(d_k) = \sum_{i=1}^4 \mathbb{P}(v_i) O_k(v_i, d_k), \quad (45)$$

where $\mathbb{P}(v_i)$ represents the probability of event v_i . That is, $\mathbb{P}(v_1) = p_L(\theta_u) p_L(\theta_k)$, $\mathbb{P}(v_2) = p_L(\theta_u) p_N(\theta_k)$, $\mathbb{P}(v_3) = p_N(\theta_u) p_L(\theta_k)$, $\mathbb{P}(v_4) = p_N(\theta_u) p_N(\theta_k)$.

For each event v_i , $O_k(v_i, d_k)$ can be derived as in the following:

(a) **Event v_1** : Using (41) and (43) in (37), we have

$$\begin{aligned} O_k(v_1, d_k) &= 1 - \left(\sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} f_u(m, \ell) (\Gamma'_u)^m e^{-b_u \Gamma'_u} \right) \\ &\quad \times \left(\sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} f_d(m, \ell) (\Gamma'_k d_k^{\alpha_k})^m \exp(-b_k \Gamma'_k d_k^{\alpha_k}) \right). \end{aligned} \quad (46)$$

(b) **Event v_2** : Similarly, we obtain $O_k(v_2, d_k)$ using (41) and (44) in (37). (c) **Event v_3** : $O_k(v_3, d_k)$ using (42) and (43) in (37). (d) **Event v_4** : $O_k(v_4, d_k)$ using (42) and (44) in (37).

B. Overall SNR Outage of the MultiCasting System, $O^{\text{MC}}(d_k)$

Conditioned on d_k , the overall network outage of the considered multicasting network (similar to overall BEP) can be expressed as follows:

$$O^{\text{MC}}(d_k) = \sum_{k=1}^{2^{\mathcal{K}}-1} \frac{1}{2^{\mathcal{K}}-1} \prod_{k=1}^{\mathcal{K}} (1 - O_k(d_k))^{1-s_k} (O_k(d_k))^{s_k}, \quad (47)$$

where $s_k = 1$ when the transmission fails for the k -th RX and $s_k = 0$ when the transmission is successful. When s_k is known *a priori* for each RX k based on certain QoS constraints, the overall network outage can be expressed as follows:

$$O_{s_k}^{\text{MC}}(d_k) = \prod_{k=1}^{\mathcal{K}} (O_k(d_k))^{s_k} (1 - O_k(d_k))^{1-s_k}, \quad (48)$$

where

$$s_k = \begin{cases} 1 & O_k(v_i, d_k) > \tau'_R \\ 0 & O_k(v_i, d_k) \leq \tau'_R \end{cases}, \quad (49)$$

where τ'_R is an arbitrary constraint on outage probability.

C. SNR Outage of an Arbitrary BS

Deconditioning over d_k (which in turn implies averaging over r_k and ϕ_k) yields the outage O as follows:

$$\begin{aligned} O &= 1 - \int_0^R \int_0^{2\pi} O_k(d_k) \frac{r_k}{\pi R^2} dr_k d\phi_k \\ &= 1 - \sum_{i=1}^4 \int_0^R \int_0^{2\pi} \mathbb{P}(v_i) O_k(v_i, d_k) \frac{r_k}{\pi R^2} dr_k d\phi_k. \end{aligned} \quad (50)$$

D. Overall Energy-Efficiency

Conditioned on d_k , the energy-efficiency of the considered multicasting system can be derived as follows:

$$\text{EE}^{\text{MC}}(d_k) = \sum_{s \in \mathbf{S}} \frac{R_0 \left(\mathcal{K} - \sum_{k=1}^{\mathcal{K}} s_k \right)}{(2^{\mathcal{K}} - 1) B E_{\text{tot}}(d)} O_{s_k}^{\text{MC}}(d_k), \quad (51)$$

where R_0 depicts bits/sec for each individual link, B is in Hz, $s_k = 1$ when the transmission fails for the k -th RX and $s_k = 0$ when the transmission is successful. Note that the term $\frac{R_0(\mathcal{K} - \sum_{k=1}^{\mathcal{K}} s_k)}{B}$ represents the total number of successfully received bits of the overall network. On the other hand, when s_k is known *a priori* for each RX k based on certain QoS constraints, the overall network energy-efficiency can be expressed as follows:

$$\text{EE}^{\text{MC}}(d_k) = \frac{R_0 \left(\mathcal{K} - \sum_{k=1}^{\mathcal{K}} s_k \right)}{B E_{\text{tot}}(d)} O_{s_k}^{\text{MC}}(d_k), \quad (52)$$

where s_k is defined in (49).

E. Approximations for Rician Fading Channels

The exact end-to-end outage for Rician fading can be simplified by approximating the Bessel function of first kind in (4) using sum of pure finite real exponent series approximation $I_0(x) = \sum_{i=1}^4 S_i \exp(T_i x)$. The PDF of the non-central chi square distribution can thus be simplified as follows [45]:

$$f_X(x) = \frac{K+1}{\Omega} e^{-K - \frac{(K+1)x}{\Omega}} \sum_{i=1}^4 S_i e^{T_i \left(2\sqrt{\frac{K(K+1)x}{\Omega}} \right)}. \quad (53)$$

Subsequently, we derive the approximate CDF as follows.

Proposition 3 Approximate CDF: Integrating (53) yields the approximate CDF of the non-central chi square distribution as:

$$F_X(\Gamma) \approx \sum_{i=1}^4 S_i \left(1 - e^{-b\Gamma + 2\sqrt{b\Gamma}\eta_{2i}} + \xi_i + \psi_i \text{erf}[\sqrt{b\Gamma} - \eta_{2i}] \right) e^{-K},$$

where $\psi_i = e^{\eta_{2i}^2} \sqrt{\pi} \eta_{2i}$, $\xi_i = \psi_i \text{erf}(\eta_{2i})$, and $\eta_{2i} = \sqrt{K} T_i$. Subsequently, using **Proposition 3**, we obtain $F_{X_u}^L(\Gamma'_u)$ by substituting b with b_u , K with K_u , and Γ with Γ'_u . Also, we approximate $F_{X_k}(\Gamma'_k|d_k)$ by substituting b with b_k and K with K_k and Γ with $\Gamma'_k d_k^{\alpha_k}$, respectively, as follows:

$$\begin{aligned} F_{X_u}^L(\Gamma'_u) &\approx \sum_{i=1}^4 S_i e^{-K_u} \left(1 - e^{(-b_u \Gamma'_u + 2\sqrt{b_u \Gamma'_u} \eta_{2i})} \right. \\ &\quad \left. + \xi_i + \psi_i \text{erf}(\sqrt{b_u \Gamma'_u} - \eta_{2i}) \right), \end{aligned} \quad (54)$$

$$\begin{aligned} F_{X_k}^L(\Gamma'_k|d_k) &\approx \sum_{i=1}^4 S_i e^{-K_k} \left(1 - e^{(-\eta_1^2 d_k^{\alpha_k} + G_i d_k^{\alpha_k/2})} \right. \\ &\quad \left. + \xi_i + \psi_i \text{erf}(\eta_1 d_k^{\alpha_k/2} - \eta_{2i}) \right), \end{aligned} \quad (55)$$

where $\psi_i = e^{\eta_{2i}^2} \sqrt{\pi} \eta_{2i}$, $\xi_i = \psi_i \text{erf}(\eta_{2i})$, and $\eta_{2i} = \sqrt{K_u} T_i$, $\eta_1 = \sqrt{b_d \Gamma'_k}$ and $G_i = 2\eta_{2i} \eta_1$. Also S_i and T_i are in vector form. $\mathbf{S} = [0.1682, 0.1472, 0.445, 0.2382]$, $\mathbf{T} = [0.7536, 0.9739, -0.715, 0.2343]$. Next, we approximate (55) as shown in (56), shown at the bottom of this page, where (56)(a) follows from the approximate CDF in (55) referred to as *Four-Node Approximation*, (56)(b) follows from the proposed approximation of $\text{erf}(x)$ as:

$$\text{erf}(x) = 1 - a_1 e^{-b'x^2} - a_2 e^{-2b'x}, \quad (57)$$

where $E_1 = e^{-K_k} \sum_{i=1}^4 S_i (1 + \xi_i + \psi_i)$ and $\mathbf{a} = [0.3017 \ 0.4389]^T$, and $b' = 1.051$. We refer this approximation as *Four-Node with Error Function Approximation*. Note that (57) is different from the approximation in [26], i.e.

$$\text{erf}(x) \approx 1 - \sum_{n=1}^{N=2} a_n e^{-nb'x^2} \approx 1 - a_1 e^{-b'x^2} - a_2 e^{-2b'x^2}. \quad (58)$$

Then, (56)(c) follows from truncating the last two terms of the *Four-Node approximation*. We refer this approximation as *Two-Node with Error Function Approximation*. Note that this is different from finding the two-node approximation of the Bessel function to find the density function. *The applications of these approximations are discussed in the following Remark and in the next section.*

$$\begin{aligned} F_{X_k}^L(\Gamma'_k|d_k) &\stackrel{(a)}{=} \sum_{i=1}^4 S_i e^{-K_k} \left(1 - e^{(-\eta_1^2 d_k^{\alpha_k} + G_i d_k^{\alpha_k/2})} + \xi_i + \psi_i \text{erf}(\eta_1 d_k^{\alpha_k/2} - \eta_{2i}) \right) \\ &\stackrel{(b)}{=} 1 - E_1 + e^{-K_k} \sum_{i=1}^4 S_i \left\{ e^{(-\eta_1^2 d_k^{\alpha_k} + G_i d_k^{\alpha_k/2})} + a_1 \psi_i e^{-b'(\eta_1 d_k^{\alpha_k/2} - \eta_{2i})^2} + a_2 \psi_i e^{-2b'(\eta_1 d_k^{\alpha_k/2} - \eta_{2i})} \right\} \\ &\stackrel{(c)}{=} 1 - E_1 + e^{-K_k} \sum_{i=1}^2 S_i \left\{ e^{(-\eta_1^2 d_k^{\alpha_k} + G_i d_k^{\alpha_k/2})} + a_1 \psi_i e^{-b'(\eta_1 d_k^{\alpha_k/2} - \eta_{2i})^2} + a_2 \psi_i e^{-2b'(\eta_1 d_k^{\alpha_k/2} - \eta_{2i})} \right\}. \end{aligned} \quad (56)$$

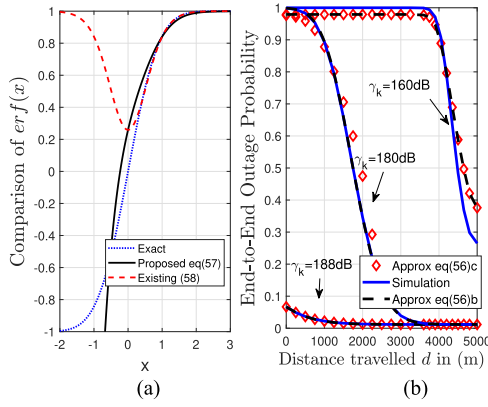


Fig. 3. Validation of (a) $\text{erf}(x)$ and (b) outage probability of RX k with Rician K-factor $K = 10$, $\bar{\gamma}_u = 172$ dB with $\tau = 0.3$.

Remark: The proposed approximations of the Rician CDF listed in (56)(a), (56)(b), and (56)(c) can be directly used to compute the end-to-end outage probability expressions derived in Section IV.

Figure 3(a) validates the proposed approximations in (56) for the exact $\text{erf}(x)$. The existing approximation in (58) provides tighter bound to $\text{erf}(x)$ for $x \geq 0$. However, the proposed approximation in (56) provides a slightly loose bound which is remarkably valid for a wider region (polarity of the approximation remains the same as $\text{erf}(x)$ function). Figure 3(b) compares the SNR outage using *Four-Node approximation*, *Two-Node approximation*, and simulation results. The results depict that the two-node approximation is less precise compared to the four-node approximation; however, it has significant analytical tractability.

V. UAV DISTANCE OPTIMIZATION PROBLEMS

In this section, we consider two different optimization problems to minimize the end-to-end energy consumption and outage probability of RX k located at cell-center. The objectives of the problems are (i) energy-constrained SNR outage minimization and (ii) outage-constrained energy consumption minimization. Note that $d_k = \sqrt{(D-d-r_k \cos(\theta'_k))^2 + (r_k \sin(\theta'_k))^2 + H^2}$. However, in this section, we consider that serving region (i.e. the cell radius R) in which all receivers are located is very small compared to the traveling distance of UAV $r_k = 0$, i.e. $d_k = d_0 = \sqrt{(D-d)^2 + H^2}$ for simplicity. Under this assumption, the farthest or nearest receiver becomes approximately the same as the receiver who is located at the cell-center. Also, we consider K and α as independent of elevation angles. We demonstrate in the results section that the variation in elevation angle during the UAV travel does not significantly impact α and K .

A. Outage-Constrained Energy Minimization

We minimize energy consumption $E_{\text{tot}}(d)$ in (14) under the SNR outage constraints as formulated in the following:

$$\begin{aligned} \mathbf{P1} : & \min_d E_{\text{tot}}(d) \\ \text{s.t.} \quad & \mathbf{C1} : O_k(v_i, d_0) \leq \tau, \\ & \mathbf{C2} : d \leq D, \quad d \geq 0, \end{aligned}$$

where $E_{\text{tot}}(d) = \frac{(P_h + p_d)\sqrt{(D-d)^2 + H^2}}{c} + \frac{P(V)d}{V} + C$ is convex. The convexity is verified by noting that the second derivative $E_{\text{tot}}''(d) \geq 0$ is non-negative with respect to optimization variable d , hence $E_{\text{tot}}(d)$ is a convex function.

Optimization for Events v_1 and v_3 : First we solve **P1** considering downlink LoS Rician fading channel. This solution approach will be applicable for the events v_1, v_3 for which

$$O_k(v_i, d_0) = 1 - U \sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} f_d(m, \ell) (\Gamma'_k d_0^\alpha)^{m_2} e^{(-b_d \Gamma'_k d_0^\alpha)}. \quad (59)$$

For event v_1 , $U = (1 - F_{\gamma_u}^L(\Gamma_0))$ can be calculated from (41). For event v_3 , $U = (1 - F_{\gamma_u}^N(\Gamma_0))$ can be calculated from (42). The outage probability constraint **C1** is complicated because of the infinite double summations. To simplify **C1**, we use (55) since the approximation is analytically tractable. However, (i) the approximation still has the error function $\text{erf}(\cdot)$ with the optimization variable d inside it, and (ii) the weights T_i in (55) are mixed (both positive and negative) and we know that “non negative weighted sum of convex function is convex”. As such, obtaining a convex function is a problem due to T_3 .

To resolve (i) and (ii), we use (56)(b) and (56)(c) approximations, respectively. The modified problem **P1** be given by substituting $O_k(v_i, d_k)$ in **C1** as follows:

$$O_k(v_i, d_0) = 1 - U \left(1 - E_1 + e^{-K} \sum_{i=1}^2 S_i \left\{ e^{(-\eta_1^2 d_0^\alpha + G_i d_0^{\alpha/2})} + a_1 \psi_i e^{-b'(\eta_1 d_0^{\alpha/2} - \eta_2 i)^2} + a_2 \psi_i e^{-2b'(\eta_1 d_0^{\alpha/2} - \eta_2 i)} \right\} \right).$$

Then, we transform the optimization variable $z = d_0^{\alpha/2} = ((D-d)^2 + H^2)^{\alpha/4}$ to simplify **C1** as follows:

$$O_k(v_i, z) = 1 - U \left(1 - E_1 + e^{-K} \sum_{i=1}^2 s_i \left\{ e^{(-\eta_1^2 z^2 + G_i z)} + a_1 \psi_i e^{-b'(\eta_1 z - \eta_2 i)^2} + a_2 \psi_i e^{-2b'(\eta_1 z - \eta_2 i)} \right\} \right).$$

Subsequently, **P1** can be reformulated as follows:

$$\begin{aligned} \mathbf{P2} : & \min_z E_{\text{tot}}(z) \\ \text{s.t.} \quad & \mathbf{C1} : O_k(v_i, z) \leq \tau \\ & \mathbf{C2} : z \leq (H^2 + D^2)^{\alpha/4}, z \geq H^{\alpha/2}, \end{aligned}$$

where $E_{\text{tot}}(z) = \frac{(P_h + p_d)z^{2/\alpha}}{c} + \frac{P(V)(D - \sqrt{z^{4/\alpha} - H^2})}{V} + C$ is monotonically decreasing function hence quasi-convex (for proof please see **Appendix B**). Therefore, we replace $E_{\text{tot}}(z)$ in **P2** with z in **P3**. The obtained outage expression is the sum of log-concave (concave function in the argument of first and second exponents) and log-affine (due to linear argument of third exponent) functions which is non-convex. To write the function in the form of log of sum of exponential functions, we use the binomial approximation $(1+x)^n \approx 1 + nx$ after completing squares in the argument of first exponential. This step linearizes the arguments of the three

exponents which makes **C1** log-affine. **P3** can thus be written as follows:

$$\begin{aligned} \mathbf{P3} : \min_z \quad & z \\ \text{s.t. } \mathbf{C1} : \quad & \sum_{i=1}^2 S_i \left(e^{\eta_1^2(2z - \frac{G_i}{2\eta_1^2})} g + a_1 \psi_i e^{-b' \eta_{2i}^2 (\frac{2\eta_1}{\eta_{2i}} z - 1)} \right. \\ & \left. + a_2 \psi_i e^{-2b'(\eta_1 z - \eta_{2i})} \right) \geq \tau' \\ \mathbf{C2} : \quad & z \leq (H^2 + D^2)^{\alpha/4}, z \geq H^{\alpha/2}, \end{aligned}$$

where $\tau' = e^K \left(\frac{\tau-1}{U} + 1 - E_1 \right)$. Note that **P3** is now convex with **C1** in the form of sum of log-affine exponential function [46] which is convex since the weights are positive and argument within exponential functions are linear and can be written as log-sum-exponents. **P3** can thus be solved using standard convex optimization techniques.

Optimization for Events v_2 and v_4 : For events with downlink NLoS Rayleigh fading channels, **P3** can be solved by substituting $O_k(v_i, z) = 1 - Ue^{-b_k \Gamma'_k z^2}$ using (44). Note that, for event v_2 , $U = (1 - F_{\gamma_u}^L(\Gamma_0))$ can be calculated from (41) and, for event v_4 , $U = (1 - F_{\gamma_u}^N(\Gamma_0))$ is given from (42).

Proposition 4: (Closed-Form Optimal Solution d^ for Rayleigh Fading Downlink Channels) Since the objective is monotonically decreasing with respect to z , the solution lies at the boundary where the equality of **C1** of **P3** holds. The closed-form optimal solution z^* can be obtained by solving the constraint $O_k(v_i, z) = 1 - Ue^{-b_k \Gamma'_k z^2} = \tau'$. By ignoring the positive root which is infeasible as it results in optimal value out of the range of d , we obtain d^* given as follows:*

$$d^* = D - \sqrt{\left(\frac{\Omega_u}{\Gamma'_u} \log \left(\frac{1-\tau}{U} \right) \right)^{2/\alpha} - H^2}. \quad (60)$$

In addition, $d^* = 0$ and $d^* = D$, if the constraint satisfies at $d = 0$ and $d = D$, respectively.

B. Energy-Constrained SNR Outage Minimization

Here, we minimize SNR outage subject to energy constraints $E_{tot}(z) \leq E_\tau$ such that the total energy consumption remains within minimum energy budget of UAV. We formulate the problem using (38) and (14) in variable z as follows:

$$\begin{aligned} \mathbf{P1} : \min_z \quad & O_k(v_i, z) \\ \text{s.t. } \mathbf{C1} : \quad & \frac{(P_h + p_d)z^{2/\alpha}}{c} + \frac{P(V)(D - \sqrt{z^{4/\alpha} - H^2})}{V} \\ & + C \leq E_\tau, \\ \mathbf{C2} : \quad & z \leq (H^2 + D^2)^{\alpha/4}, z \geq H^{\alpha/2}, \\ \mathbf{C3} : \quad & \frac{m}{z^2} - b_d \Gamma'_k < 0, \end{aligned}$$

where constraint **C3** is the feasibility constraint. Under this feasibility condition, we can replace $O_k(v_i, z)$ with z in the objective function as follows:

$$\begin{aligned} \mathbf{P2} : \min_z \quad & z \\ \text{s.t. } \mathbf{C1} : \quad & \frac{(P_h + p_d)z^{2/\alpha}}{c} + \frac{P(V)(D - \sqrt{z^{4/\alpha} - H^2})}{V} \\ & + C \leq E_\tau, \\ \mathbf{C2} \& \mathbf{C3}. \end{aligned}$$

Under the feasibility regime, the outage probability $O_k(v_i, d_k) \forall i = 1, 2, 3, 4$ is monotonically decreasing function of z (see **Appendix C** for proof). Due to the monotonic nature of the objective and constraint, the solution to the above problem lies at the boundary where **C1** becomes an equality constraint.

Proposition 5 (Closed-Form Optimal UAV Distance for Energy-Constrained Outage Minimization): The optimal solution z^ can be obtained by solving $E_{tot}(z) - E_\tau = 0$, taking the solution with positive root and then substituting z^* in $z = d_0^{\alpha/2}$ to obtain d^* by taking the solution with negative root:*

$$d^* = D - 2\sqrt{\frac{(\kappa_3 + F^2 + 2F\sqrt{\kappa_3})}{(\kappa_1^2 - 1)^2} - H^2}, \quad (61)$$

where $\kappa_3 = \kappa_1^2(F^2 + \kappa_2^2)$, $\kappa_2^2 = (\kappa_1^2 - 1)H^2$, $\kappa_1 = \frac{cP(V)}{(P_h + p_d)V}$ and $F = \left(E_\tau - C - D \frac{P(V)}{V} \right) \frac{c}{(P_h + p_d)}$. This solution is valid for events v_i , $\forall i = 1, \dots, 4$.

C. Multi-Objective Optimization

In this subsection, we simultaneously minimize both the energy consumption and outage probability. We formulate a bi-objective optimization problem with weights λ and $1 - \lambda$ to the energy consumption and outage probability, respectively. The problem is formulated as follows:

$$\begin{aligned} \mathbf{P1} \quad & \min_z \lambda E_{tot}(z) + (1 - \lambda) O_k(z) \\ \text{s.t. } \mathbf{C1} : \quad & O_k(v_i, z) \leq \tau \\ \mathbf{C2} : \quad & z \leq (H^2 + D^2)^{\alpha/4}, z \geq H^{\alpha/2}. \end{aligned} \quad (62)$$

In (62) the unit of $E_{tot}(z)$ is Joules and of the order of thousands, whereas the outage probability is unitless quantity between '0' to '1'. The objective function is implicitly biased towards energy consumption. With the help of normalization of $E_{tot}(z)$ (i.e. shifting and scaling of $E_{tot}(z)$ as $\frac{E_{tot}(z) - E_{\min}}{E_{\max} - E_{\min}}$), (i) we reduce the magnitude of $E_{tot}(z)$ from '0' to '1' which was required to create a balance between the energy consumption and outage probability, and (ii) we make $E_{tot}(z)$ a unitless quantity. Now both the terms are of same order and can be added together. Note that $E_{\max}$ and $E_{\min}$ represent maximum and minimum value of energy consumption that is obtained by substituting the limits of $z = H^{\alpha/2}$ and $z = (D^2 + H^2)^{\alpha/4}$, respectively, in $E_{tot}(z)$ defined in Section V.B problem **P1**. That is, $E_{\max} = \frac{(P_h + p_d)H}{c} + \frac{P(V)}{V}D + C$ and $E_{\min} = \frac{(P_h + p_d)\sqrt{H^2 + D^2}}{c} + C$. The objective of **P1** can thus be rewritten as follows:

$$U(z) = O_k(z) + \bar{\nu} (E_{tot}(z) - E_{\min}(z)), \quad (63)$$

where $\bar{\nu} = \frac{\lambda}{1-\lambda} \frac{1}{E_{\max} - E_{\min}}$ minimizes outage probability when $\frac{\lambda}{1-\lambda} = 0$ and minimizes energy consumption when $\frac{\lambda}{1-\lambda} = \infty$ [47], [48]. The selection of weights is a choice of decision maker. To solve **P1**, we use the fact that the objective function can be formulated as a difference of two convex functions as follows:

$$U(z) = \bar{\nu} (E_{tot}(z) - E_{\min}) - (-O_k(z)). \quad (64)$$

Algorithm 1 DC Algorithm

Data: Initial point: z^0 , Max Iterations $J_{\max}$, Error tolerance

ϵ , Objective function $Q(z)$ and reduction of candidate

points based on $0 \leq O_k(z) \leq \tau$

Result: sub optimal desired solution z_{sub}

set $j = 0$; **while** $|Q(z^{j+1}) - Q(z^j)| \leq \epsilon$ **or** $j > J_{\max}$ **do**

- Convex approximation of $Q(x)$ at the point z^j ,
 $Q^j(z) = G(z) - H(z^j) - H'(z^j)(z - z^j)$
- Solve $z^j = \arg\min Q^j(z)$
- $j = j + 1$.

end

By defining $Q(z) = G(z) - H(z)$ where $G(z)$ and $H(z)$ are convex functions given by $G(z) = \bar{\nu}(E_{tot}(z) - E_{min}(z))$ and $H(z) = -O_k(z)$, respectively, we formulate **P1** as follows:

$$\begin{aligned} \mathbf{P2} \quad & Q(z) = \min_z G(z) - H(z) \\ \text{s.t.} \quad & \mathbf{C1} : O_k(v_i, z) \leq \tau \\ & \mathbf{C2} : z \leq (H^2 + D^2)^{\alpha/4}, z \geq H^{\alpha/2}, \end{aligned} \quad (65)$$

where

$$\begin{aligned} G(z) &= \bar{\nu} \left(\frac{(P_h + p_d)}{z^{-2/\alpha_c}} + \frac{P(V)(D - \sqrt{z^{4/\alpha} - H^2})}{V} + C - E_{min}(z) \right), \\ H(z) &= \max \left\{ -1 + U \left(1 - E_1 + e^{-K} \sum_{i=1}^2 S_i \left\{ e^{\eta_1^2(2z - \frac{\eta_2 i}{\eta_1})} g \right. \right. \right. \\ &\quad \left. \left. \left. + a_1 \psi_i e^{-b_n \eta_{2i}^2 (\frac{2\eta_1}{\eta_{2i}} z - 1)} + a_2 \psi_i e^{-2b_n(\eta_1 z - \eta_{2i})} \right\} \right), 0 \right\}. \end{aligned}$$

It should be noted that the outage probability approximation results in the value beyond $[0 \ 1]$ range therefore to limit the range we use $\max(\cdot, 0)$. It is clear from **Appendix A** that $G(z)$ is convex and $H(z)$ is also convex, respectively, with respect to z . However, in general, (65) is not a convex/concave problem. That is, the objective function is a difference of convex functions. DC programming approach provides an efficient way to get local optima numerically which occasionally turns out to be global optimal [49], [50]. To find the efficient sub-optimal solution, this method involves successive convex approximation. That is, the sub-problem in each iteration j is obtained by linearization of $H(z)$ at point z^j , i.e. $H(z) = H(z^j) + H'(z^j)(z - z^j)$ where $H'(\cdot)$ represents the derivative of $H(\cdot)$. Note that z^j is update in z after j -th iteration. The algorithm to solve **P2** is presented in **Algorithm 1**. The complexity and convergence of **Algorithm 1** is well-established in [51].

VI. NUMERICAL RESULTS AND DISCUSSION

We verify the accuracy of our derived expressions and obtain insights related to the optimal ferrying distance, and performance of a UAV-assisted data ferrying distance system

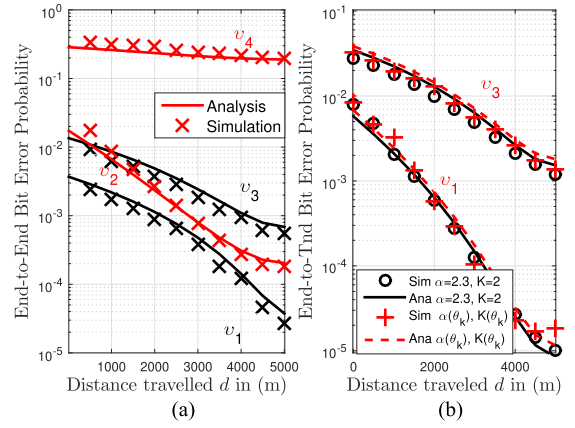


Fig. 4. (a) BEP conditioned on d_k , $\gamma_u = 211\text{dB}$, $\gamma_k = 181\text{dB}$, $\Gamma_0 = 10\text{dB}$, $K(\theta_k)$ and $\alpha(\theta_k)$, (b) BEP with $K(\theta_k)$, ranging from $[2.18 \ 1.92]$ and $\alpha(\theta_k)$ ranging from $[2.313 \ 2.311]$ vs fixed K, α .

in a variety of scenarios. Unless stated otherwise, the simulation parameters are $D = 5000\text{m}$, $H = 500\text{m}$, $p_U = p_D = 50\text{W}$, $\eta_1 = 0.009$, $\eta_D = 0.01$, $R = 200\text{m}$, $\Gamma_0 = 10\text{dB}$, $q_U = q_D = -1.5$, $v_U = v_D = 3.5$, $w_U = w_D = 15\text{dB}$, $z_U = z_D = 5$. The typical $N_0 = 10^{-17}\text{W/Hz}$ [52], that justifies the values we used for γ_u and γ_k herein.

Figure 4(a) presents the end-to-end BEP of a typical RX as a function of the ferrying distance d considering the events v_1 , v_2 , v_3 , and v_4 separately that appear in the order from bottom to top. The BEP decreases monotonically as a function of d . Further, the theoretical results match well with the simulation results. Also, we observe that v_1 (Rician (LoS)-only) and v_4 (Rayleigh (NLoS)-only) have the minimum and maximum BEPs, respectively. The BEP in other events is similar to v_1 which shows that the significance of LoS transmissions in any one hop has a positive impact on the BEP.

Figure 4(b) depicts the comparison of the end-to-end BEP of two events v_1 and v_3 for Rician K -factor and path-loss exponent α as a function of elevation angle θ_k as in (6) and (7) to the fixed K and α . We note that the change in ferrying distance does not impact θ_k and in turn $K(\theta_k)$, $\alpha(\theta_k)$ significantly. Thus the end-to-end BEP remains the same. This result also justifies our assumption to consider fixed value of K and α in the optimization problems presented in Section V. Note that the fixed K and α can not be chosen randomly but their value needs to be chosen close to $K(\theta_k)$, $\alpha(\theta_k)$.

Figure 5 depicts the end-to-end PEP as a function of d without and with ARQ retransmissions. Analytical and simulation results match well for $n = 0$, i.e. for single transmission and $n = 1, 2, 3$ with two and three retransmissions, respectively. As expected, PEP decreases with d which means reliability is increasing. Figure 5 depicts that increasing retransmissions help in achieving the desired reliability with a lower traveled distance which will minimize the energy consumption. In addition, we also note that the reduction in PEP is significant $n = 0$ to $n = 1$ and the degree of reduction gradually reduces for $n = 2$ to $n = 3$ and onwards.

Figure 6 depicts the end-to-end outage probability of a typical RX as a function of the distance traveled by the UAV. Each curve corresponds to the events v_1 , v_2 , v_3 , and v_4

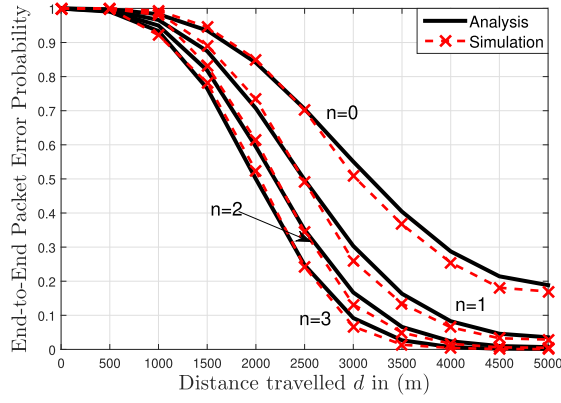


Fig. 5. PEP conditioned on d_k : comparison of simulation and exact analysis for unconditioned on events and conditioned on d_k as a function d with and without ARQ.

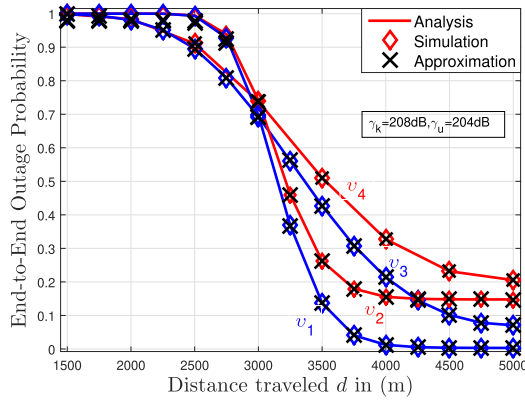


Fig. 6. Outage probability conditioned on d_k comparison of simulation, exact, and approximate analysis for different d .

separately. Theoretical results match well with the simulations. Figure. 6 demonstrates the efficacy of the Bessel approximation. The outage probability decreases monotonically as a function of the ferrying distance d due to reduction in path-loss. As such, d can be optimized to meet a specific outage probability constraint.

We note that v_1 (Rician-only) and v_4 (Rayleigh-only) shows the minimum and maximum outages, respectively, beyond a certain travel distance $d \approx 3500$ m. For distances smaller than $d = 3500$ m, Rician-only event v_1 (both uplink and downlink are LoS) performs similar to Rician-Rayleigh event v_2 (uplink is LoS and downlink is NLoS). The reason is the dominance of uplink over downlink due to reduced path-loss in the uplink (when the UAV is close to $\mathbf{D}$). Similarly, Rayleigh-only event v_4 (both uplink and downlink channels are NLoS) performs similar to Rayleigh-Rician event v_3 (uplink is NLoS and downlink is LoS). For distances more than $d = 4250$ m, Rician-only event v_1 performs similar to Rayleigh-Rician event v_3 and Rayleigh-only event v_4 performs similar to Rician-Rayleigh event v_2 . The reason is the dominance of downlink over uplink transmission when the UAV is close to $\mathbf{F}$.

Figure 7 depicts the end-to-end outage probability of a typical RX as a function of the altitude taken by the UAV considering different values of e_k as given in (3). The derived theoretical results match well with the Monte-Carlo simula-

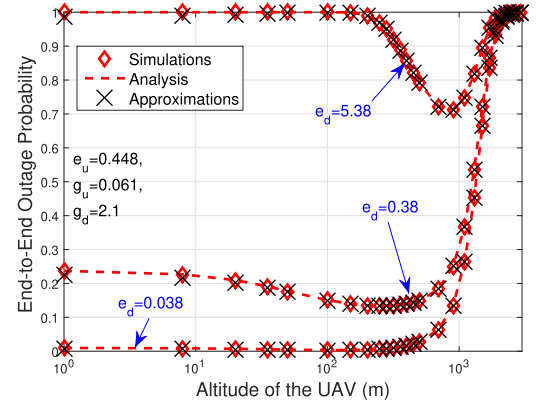


Fig. 7. Outage probability as a function of the altitude of UAV considering $\gamma_u = 172$ dB, $\gamma_k = 184$ dB, and $d = 2000$ m.

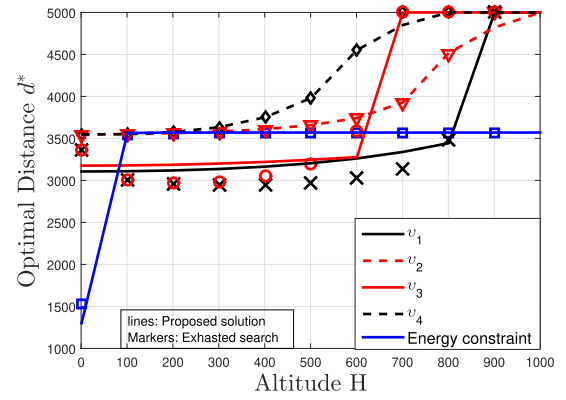


Fig. 8. Optimal distance d^* plotted for (a) outage constrained energy minimization (black and red) and (b) energy constrained outage minimization (blue), $\gamma_u = 188$ dB, $\gamma_k = 172$ dB, & $K = 5.25$ with outage threshold $\tau = 0.2$, and energy budget $E_T = 1 \times 10^5$.

tion results. As e_k decreases, the LoS probability increases (see (3)) and therefore the end-to-end outage decreases. Another interesting trend is that the SNR outage initially decreases with the increase in height (due to higher LoS probability); however, after a certain point the SNR outage tends to increase due to the increasing distance from the ground. It is also interesting to observe that the reduction in outage due to higher altitude is noticeable only for higher values of e_k . The energy efficiency curve shows the inverse trend of outage probability.

Figure 8 shows the optimal distance d^* obtained by solving the optimization problems (i) outage constrained energy minimization and (ii) energy constrained outage minimization, for all four events v_1, v_2, v_3 , and v_4 as a function of the UAV altitude H . We note that the optimal ferrying distance d^* in the former problem gradually increases with the height of UAV for all events. The reason is that, with the increase of height, the channel becomes weaker due to path-loss dominance which causes UAV to travel more to meet the constraint on outage probability. For the latter problem, d^* increases first and then becomes a constant almost at $d^* = 3500$. The derived optimal solutions match well with the optimal results obtained through exhaustive search. We also note that the optimal distance for the event v_1 (Rician-only) is minimum because of improved channel condition. However, v_4 (Rayleigh-only)

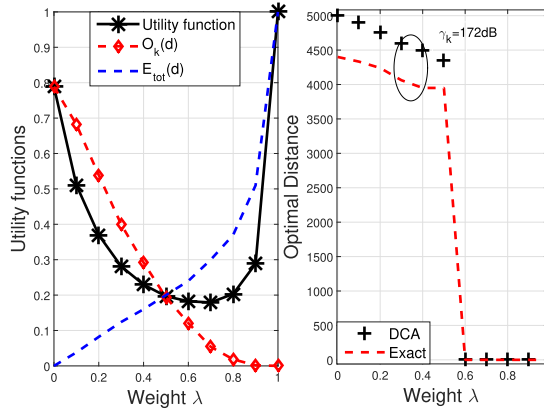


Fig. 9. Multiobjective optimization with minimization of outage and energy consumption for a typical user for Rician K -factor $K = 2$ for event v_1 , $\gamma_u = 176\text{dB}$ and $\gamma_k = 180\text{dB}$ as a function of weights λ .

suffers with worst channel attenuation which causes the d^* to be the longest. The optimal distances of the v_2, v_3 lies between the v_1 and v_4 .

Figure 9(a) shows the impact of λ on the utility function in (64). For smaller values of λ , outage probability is dominant in the utility function. As λ increases, the bias towards energy consumption increases and at $\lambda = 0.5$ both outage probability and energy consumption are equally important. Figure 9(b) depicts the comparison of the optimal ferrying distance using exhaustive search to the optimal solution obtained from **Algorithm 1** with $0 \leq O_k(d_k) \leq 0.5$. In the region for $\lambda \geq 0.6$, the objective is biased towards minimizing the energy consumption which in turn minimizes the UAV distance traveled while meeting the outage constraints. With no outage constraints, the energy is minimum when UAV does not travel from **I**. In the region for $\lambda \leq 0.6$, the objective is biased towards minimizing the outage which in turn maximizes the UAV distance traveled while meeting the minimum energy consumption constraints. With no energy constraints, the outage is minimum when UAV is at **F**.

A summary of the major observations is as follows: The optimal data ferrying distance d^* gradually increases with the height of UAV since with the increase of height the channel becomes weaker and, in turn, the UAV needs to travel more to meet the constraint on outage probability. Increasing retransmissions help in achieving the desired reliability with a lower data ferrying distance. This minimizes the energy consumption. In addition, a reduction in packet error probability is higher with one retransmission and it gradually decreases for a higher number of retransmissions. Regarding the impact of weight λ on the utility function in our multi-objective optimization problem, for $\lambda = 0$, the outage probability is dominant; therefore, the UAV tends to travel a larger distance in order to minimize outage. When $\lambda = 1$, the bias towards energy consumption increases due to which the UAV tends to minimize its travel distance to minimize energy consumption. For any other value of λ , the UAV finds optimal distance somewhere between 0 to D to fulfill the outage-energy trade-off.

VII. CONCLUSION

We have characterized the end-to-end performance in terms of transmission outage and bit error probability of a UAV-assisted ferrying network with multi-casting. We have modeled LoS and NLoS transmission links with Rician and Rayleigh fading channels, respectively. We have provided tractable approximation of Rician channel statistics. We have formulated and solved the outage-constrained energy minimization and energy-constrained outage minimization problems to optimize the ferrying distance. We have convexified the two problems to obtain optimal solutions and have shown that the obtained optimal solutions match well with the solutions obtained from exhaustive search.

This work can be extended for a more general UAV trajectory where a UAV may have to serve multiple clusters on ground. That is, by iterating our proposed optimal algorithm such that all the clusters meet their QoS one after the other. The steps include: (1) Finding the nearest UAV cluster from the initial UAV location, where the nearest cluster is obtained by comparing the initial UAV location to the cell center of all clusters which are not yet served; (2) Finding the optimal hovering location with respect to the nearest cluster to be served; (3) Taking the optimal UAV location as the initial UAV location, now find the next nearest cluster, as given in the first step; (4) Finding the optimal UAV travel distance for the next cluster to be served, and so on. Therefore, application of the proposed model for different trajectories and studying the trade-off between retransmissions and energy consumption can be potential future directions.

APPENDIX A: CONVEXITY OF $E_{tot}(z)$

The second derivative of $E_{tot}(z)$ is $\frac{2}{\alpha} z^{\frac{2}{\alpha}-2} \left[\frac{(P_h + p_d)}{c} \left(\frac{2}{\alpha} - 1 \right) + \frac{A}{H^2} z^{4/\alpha} \left(z^{-2} - \frac{2}{\alpha} \right) + \left(\frac{4}{\alpha} - 1 \right) A \right]$, where $A = \frac{P(v) z^{2/\alpha} H^2}{(z^{4/\alpha} - H^2)^{3/2}}$. The convexity is not conclusive from the second derivative for all values of z . However, we verify the convexity on MATHEMATICA and CVX for different values of H , D and α . We can check the convexity of $E_{tot}(z)$ in CVX by writing the following problem:

$$\begin{aligned} \min_z \quad & E_{tot}(z) = (P_h + p_d) \frac{z^{2/\alpha}}{c} \\ & + \frac{P(V)(D - \sqrt{z^{4/\alpha} - H^2})}{V} + C \\ \text{s.t.} \quad & z \leq (H^2 + D^2)^{\alpha/4}, z \geq H^{\alpha/2} \end{aligned}$$

in CVX, using auxiliary variable W , as shown below:

$$\begin{aligned} \min_z \quad & \frac{(P_h + p_d) z^{2/\alpha}}{c} + \frac{P(V)(D - W)}{V} \\ \text{C3:} \quad & (z^{2/\alpha} - H)(z^{2/\alpha} + H) \geq W^2 \\ \text{C2:} \quad & z \leq (H^2 + D^2)^{\alpha/4}, z \geq H^{\alpha/2} \end{aligned}$$

which is convex and solvable in CVX.

APPENDIX B: MONOTONICITY OF $E_{tot}(z)$

We demonstrate that $E_{tot}(z)$ is monotonically decreasing for all values of α , H , and D given $P_h, P(V), P_c$. We first

substitute $y = z^{2/\alpha}$ and show that $E_{tot}(y)$ is monotonically decreasing if $E'_{tot}(y) = \frac{(P_h + p_d)}{c} - \frac{P(V)}{V} \frac{y}{\sqrt{y^2 - H^2}} \leq 0$. That is, $\frac{(P_h + p_d)V}{cP(V)} \leq \frac{y}{\sqrt{y^2 - H^2}}$. Since $(P_h + p_d) < P(V)$, V is of order of thousands, c is in order of 10^8 , the ratio $\frac{V}{c}$ is always less than unity with order 10^{-5} . The $(P_h + p_d)V < P(V)c$. Therefore, L.H.S. is upper bounded by 1. On the other hand, we note that the R.H.S. $\frac{y}{\sqrt{y^2 - H^2}} > 1$. Starting from $y = z^{2/\alpha} = H$, we see that the R.H.S becomes infinity. Then, for $y = H + 1$, we see that the R.H.S becomes $\frac{H+1}{\sqrt{2H+1}}$ for all y , which is greater than 1. Then, we substitute $y = H + n$, where $n \geq 2$ which gives $\frac{H+n}{\sqrt{2H+n}} > 1$. Therefore, L.H.S. < R.H.S. and the function is monotonically decreasing.

APPENDIX C: MONOTONICITY OF OUTAGE PROBABILITY

To prove outage probability for v_1 (46) and v_3 is an increasing function w.r.t. z , we take the derivative as follows:

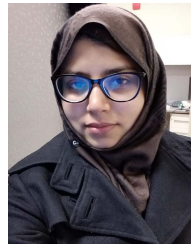
$$\begin{aligned} \frac{dO_k(v_i, z)}{dz} \\ = -U \sum_{\ell=0}^{\infty} \sum_{m=0}^{\ell} f(m, \ell) 2z^{2m+1} e^{-b_d \Gamma'_k z^2} \left(\frac{m}{z^2} - b_k \Gamma'_k \right), \\ i = 1, 3. \end{aligned}$$

That is, (46) is as an increasing function of z if $\frac{dO_k(v_i, z)}{dz} \geq 0$ which is true if $\left(\frac{m}{z^2} - b_k \Gamma'_k \right) < 0$. This is due to the reason that in $\frac{dO_k(v_i, z)}{dz}$ functions, e.g. $f(m, \ell)$, z^{2m+1} , and $e^{-b_d \Gamma'_k z^2}$ are non negative for all m , and z therefore $\left(\frac{m}{z^2} - b_k \Gamma'_k \right)$ is the only term that can change sign. Thus the first derivative will remain positive only if $(m - b_k \Gamma'_k z^2) < 0$. Note that m is in the range $m \in [0, 20]$ for evaluations, $b_k \in \{0, \infty\}$, and the range of z is $z \in \{H^{\alpha/2} (H^2 + D^2)^{\alpha/4}\}$. Therefore, it is evident that P_c and the contribution of Γ'_k are less dominant than z^2 . Hence the condition satisfies for our range of variables.

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